

6. Maclaurin and Taylor series		
6.1	Third and higher order derivatives.	
6.2	Derivation and use of Maclaurin series.	The derivation of the series expansion of e^x , $\sin x$, $\cos x$, $\ln(1+x)$ and other simple functions may be required.
6.3	Derivation and use of Taylor series.	The derivation, for example, of the expansion of $\sin x$ in ascending powers of $(x - \pi)$ up to and including the term in $(x - \pi)^3$.
6.4	Use of Taylor series method for series solutions of differential equations.	Students may, for example, be required to find the solution in powers of x as far as the term in x^4, of the differential equation $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0, \text{ such that } y = 1, \frac{dy}{dx} = 0 \text{ at } x = 0.$